PHYSICS OF SEMICONDUCTOR

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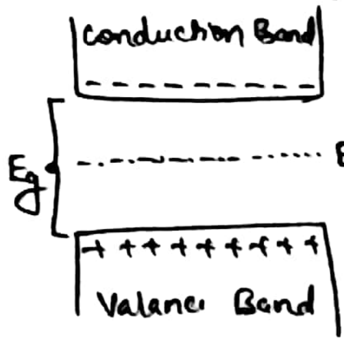
Semiconductors are those materials/Substances whose electrical conductivity/Resistivity lies between conductors and insulators. At lower temperature, semiconductors acts as a insulator and at higher tempr, they act as a conductor. The conductivity in semiconductors is due to motion of both the charge carriers ie electrons & holes. In semiconductors, the conduction band and valance band are seperated by a narrow energy gap known as Forbidden energy gap (E_g). The energy gap is nearly 1eV.

In intrinsic semiconductor, electron concentration (no of electrons) and holes concentration (no of holes) are equal. In extrinsic semiconductors, the ratio of charge carriers is not same. ie in p-type semiconductors $N_h > N_e$ and in n-type semiconductors $N_e > N_h$.

Fermi level in Intrinsic Semiconductor

Fermi energy is the average energy of conduction electrons. Fermi level is used to distinguish between filled energy levels and unfilled energy levels.



As temperature increases, electrons present at the top of valance band jump in to the bottom of conduction band due to thermal excitation. These electrons return to the valance band and hence participate in excitation and deexcitation. Therefore the excitation and deexcitation energy between top of valance band and bottom of conduction band is nearly equal to $\frac{1}{2}E_g$.

$\therefore$ Fermi level lies in the mid part of the forbidden energy gap for intrinsic semiconductor. $\Rightarrow E_F = \frac{1}{2}E_g$

For intrinsic semiconductor, fermi level is uniform but for an extrinsic semiconductor, fermi level shifts and not uniform.

Expression for electrical conductivity in semiconductors

The expression for electric current in terms of drift velocity is given by $I = neAV_d \rightarrow (1)$

Where $A \rightarrow$ Area of semiconductor.

In semiconductor the conductivity is due to motion of holes and electrons.

$\therefore$ Electric current due to motion electrons is

$(1) \Rightarrow \quad I_e = n_e e A V_e \rightarrow (2)$

Where, $n_e \rightarrow$ number density of electrons

$e \rightarrow$ charge of electron

$V_e \rightarrow$ drift velocity of electron

Electric current due to motion of holes is

$(1) \Rightarrow \quad I_h = n_h e A V_h \rightarrow (3)$

Where, $n_h \rightarrow$ number density of holes

$e \rightarrow$ charge of holes

$V_h \rightarrow$ drift velocity of holes.

$\therefore$ Total current, $I = I_e + I_h$

$\therefore (2) + (3) \Rightarrow \quad I = n_e e A V_e + n_h e A V_h$

$\Rightarrow \quad I = Ae(n_e V_e + n_h V_h)$

$\frac{I}{A} = e(n_e V_e + n_h V_h) \rightarrow (4)$

Since $\frac{I}{A} = J$, current density

$\therefore (4) \Rightarrow \quad J = e(n_e V_e + n_h V_h) \rightarrow (5)$

The relation between current density (J), conductivity (σ) and applied electric field (E) is given by $J = \sigma E \rightarrow (6)$

Equating (6) & (7), $\quad \sigma E = e(n_e V_e + n_h V_h)$

$\Rightarrow \sigma = e\left(n_e \frac{V_e}{E} + n_h \frac{V_h}{E}\right) \rightarrow (7)$

In eqn (7), $\frac{V_e}{E} = \mu_e$, mobility of electrons

and $\frac{V_h}{E} = \mu_h$, mobility of holes

$\therefore (7) \Rightarrow \boxed{\sigma = e(n_e \mu_e + n_h \mu_h)} \rightarrow (8)$

For intrinsic semiconductors, $n_e = n_h = n_i$ (intrinsic charge carrier)

$\therefore (8) \Rightarrow \boxed{\sigma_i = e n_i (\mu_e + \mu_h)} \rightarrow (9).$

For P-type semiconductor, $n_h >> n_e \therefore n_e$ is neglected $\therefore (9) \Rightarrow \boxed{\sigma_p = e n_h \mu_h}$

For N-type semiconductor, $n_e >> n_h \therefore n_h$ is neglected $\therefore (9) \Rightarrow \boxed{\sigma_n = e n_e \mu_e}$

Expression for Electron concentration (N_e):

The expression for electron concentration is given by

$$N_e = \frac{4\sqrt{2}}{h^3}\left(\pi m_e^* KT\right)^{3/2} \cdot e^{\frac{E_F - E_g}{KT}}$$

Where $m_e^* \rightarrow$ effective mass electron
$h \rightarrow$ planck's constant
$K \rightarrow$ Boltzmann's constant
$T \rightarrow$ Absolute tempr
$E_F \rightarrow$ Fermi energy, $E_g \rightarrow$ Energy gap.

Expression for Holes concentration (N_h):

The expression for holes concentration is given by

$$N_h = \frac{4\sqrt{2}}{h^3}\left(\pi m_h^* KT\right)^{3/2} \cdot e^{\frac{-E_F}{KT}}$$

Where $m_h^* \rightarrow$ effective mass of holes
$h \rightarrow$ planck's constant
$K \rightarrow$ Boltzmann's constant
$T \rightarrow$ Absolute tempr
$E_F \rightarrow$ Fermi energy.

Relation between Fermi energy (E_F) and Energy gap (E_g) for intrinsic semiconductor:

In case of intrinsic semiconductors, number of electrons per unit volume in conduction band is equal to number of holes per unit volume in valance band.

$$\therefore \quad N_e = N_h$$

on substituting,

$$\frac{4\sqrt{2}}{h^3}\left(\pi m_e^* KT\right)^{3/2} \cdot e^{\frac{E_F - E_g}{KT}} = \frac{4\sqrt{2}}{h^3}\left(\pi m_h^* KT\right)^{3/2} e^{\frac{-E_F}{KT}}$$

$$\left(m_e^*\right)^{3/2} e^{\frac{E_F - E_g}{KT}} = \left(m_h^*\right)^{3/2} e^{-E_F/KT}$$

$$\Rightarrow e^{\frac{E_F - E_g}{KT} + \frac{E_F}{KT}} = \left(\frac{m_h^*}{m_e^*}\right)^{3/2}$$

$$e^{\frac{2E_F - E_g}{KT}} = \left(\frac{m_h^*}{m_e^*}\right)^{3/2}$$

Taking natural log on both sides of above Equation

$$\frac{2E_F - E_g}{KT} = \ln\left(\frac{m_h^*}{m_e^*}\right)^{3/2} = \frac{3}{2}\ln\left(\frac{m_h^*}{m_e^*}\right)$$

$$\Rightarrow E_F = \frac{3KT}{4}\ln\left(\frac{m_h^*}{m_e^*}\right)^{3/2} + \frac{E_g}{2} \rightarrow (1)$$

In Eqn (1), $m_e^* \simeq m_h^*$ for practical consideration. $\therefore$ The first term in RHS of Eqn (1) becomes zero.

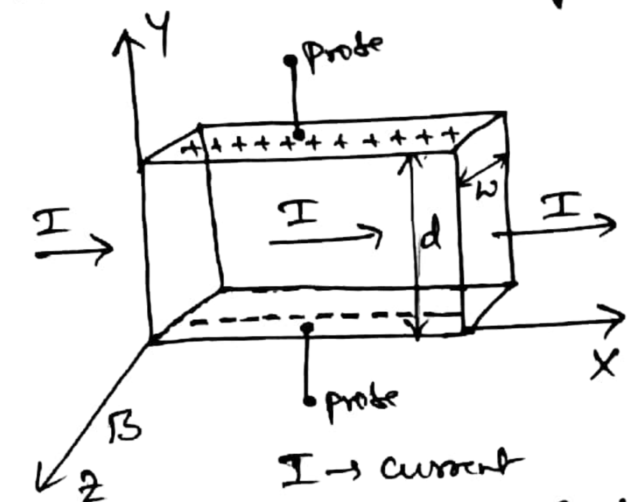
$$\therefore (1) \Rightarrow \boxed{E_F = \frac{E_g}{2}} \rightarrow (2)$$

$\therefore$ Fermi level lies exactly at the middle of energy gap.

Hall Effect and Expression for Hall Voltage (V_H) and Hall coefficient (R_H):

When a current carrying semiconductor is placed in a magnetic field, an electric field and potential difference is developed inside the material in the direction perpendicular to both magnetic field and direction of current. This effect is known as Hall effect.

Hall effect is used to find carrier concentration, mobility and type of semiconductor.



I → current

B → Magnetic field

Consider a rectangular slab of semiconductor having thickness 'd' and width 'w' is placed in a uniform magnetic field (B) acting along Z-axis. When a current (I) is passed through the semiconductor along X-axis, the charges (charge carriers) get seperated due to Lorentz force (F_L) given by the expression

$$F_L = Bqv \sin\theta \rightarrow (1)$$

$\because \theta = 90^\circ$ & $q = -e$ (for electron)

$\therefore (1) \Rightarrow F_L = -Bev \rightarrow (2)$

Where $v \rightarrow$ drift velocity

Due to Lorentz force, the negative charge experience force & hence accumilate at the bottom of the semiconductor. This can be understand by Flemings left hand rule.

Due to seperation of charges, a potential is created ($V_H \rightarrow$ Hall voltage) which can be measured using a device connected to probes as shown in figure.

The force on electron due to hall field E_H is $F_H = -eE_H \rightarrow (3)$

under equilibrium condition,

$$F_L = F_H$$

$$\Rightarrow -B\cancel{e}v = -\cancel{e}E_H$$

$$\text{Hall field, } E_H = Bv \rightarrow (4)$$

The electric field throughout the thickness of semiconductor is $E_H = \frac{V_H}{d} \rightarrow (5)$

put (5) in (4)

$(4) \Rightarrow \frac{V_H}{d} = Bv \rightarrow (6)$

current flowing through semiconductor of Area ($A = d \times w$) given current density. $\therefore J = \frac{I}{A} = \frac{I}{d \times w}$

But $I = neAv$.

$\Rightarrow J = \frac{I}{A} = nev \rightarrow (7)$

Let $ne = \rho$ charge density

$\therefore (7) \Rightarrow J = \rho v$

$\Rightarrow v = \frac{J}{\rho} \rightarrow (8)$

put (8) in (6)

$(6) \Rightarrow$ Hall voltage $V_H = Bd \cdot \frac{J}{\rho}$

$\Rightarrow V_H = B \cdot \cancel{d} \frac{I}{(w \times \cancel{d})\rho}$

$$\Rightarrow \boxed{V_H = \frac{BI}{w\rho}} \rightarrow (9)$$

But Hall field depends on J & B

$\therefore \quad E_H \propto J B$

$\Rightarrow E_H = R_H \, J B$

$\therefore \quad R_H = \dfrac{E_H}{J \cdot B} \longrightarrow (10)$

Put (4) in (10)

$(10) \rightarrow \quad R_H = \dfrac{B v}{J \cdot \cancel{B}} = \dfrac{\cancel{v}}{\rho \cancel{v}} \quad (\because J = \rho v)$

$$\boxed{R_H = \frac{1}{\rho}} \longrightarrow (11)$$

Eqn (11) is the expression for Hall coefficient.

We know, $V_H = \dfrac{B I}{W \rho} = \dfrac{B I}{W} \cdot \dfrac{1}{\rho}$

$$\boxed{V_H = R_H \cdot \frac{B I}{W}} \longrightarrow (12)$$

Eqn (12) is the expression for Hall voltage interms of Hall coefficient.

Hence, Hall coefficient is equal to reciprocal of charge density.

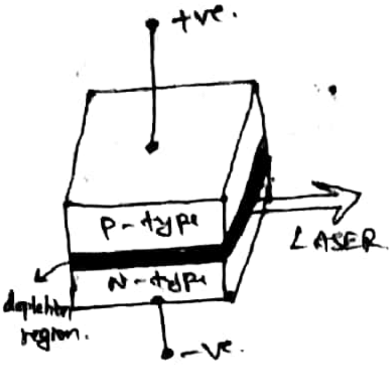
Semiconductor laser:

Principle: When a PN-junction is forward biased, across the depletion region, the conduction band electrons from N-type and holes in the valence band of P-type combine & hence radiation is emitted.

The PN-junction is heavily doped & large current is made to flow through the junction to create population inversion and hence laser can be obtained by using resonant cavity.

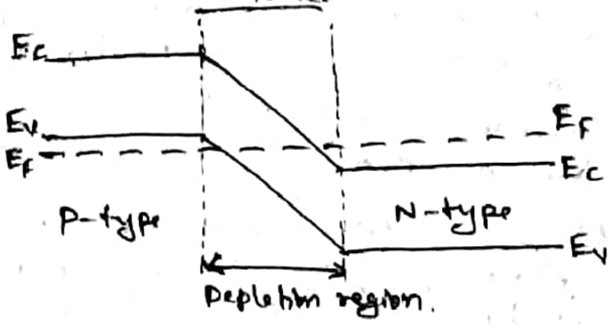
Pure semiconductors like Ge, Si emitt radiation in IR region. Doped semiconductors like Gallium Arsenide (GaAS), Gallium Arsenide phosphide (GaASP) emitts radiation in the visible region.

Construction: Metallic contacts are provided to P-type and N-type heavily doped PN-junction. Two opposite faces which are perpendicular to the plane of the junction are polished and made parallel to each other. These parallel faces constitute resonant cavity & laser is obtained from these face as shown in figure.



Energy level diagram:

Unbiased

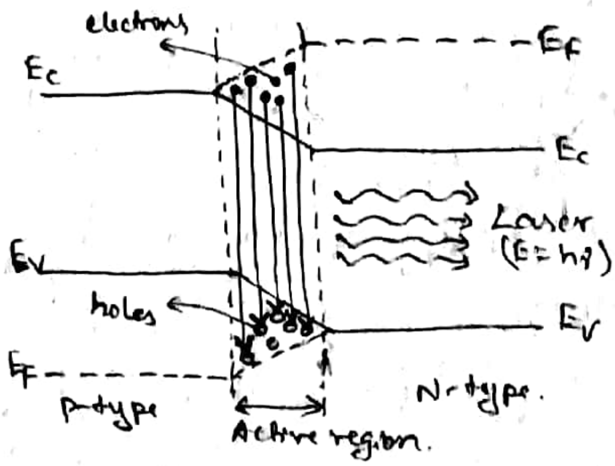


Working: In unbiased condition, the fermi level (E_F) is uniform throughout valence band (Ev) and conduction band (Ec) energy level of P-type & N-type semi conductor.

When P-N junction is forward biased, the Fermi level shifts (as shown) and width of depletion region decreases due to injuction of electrons and holes. At lower forward current, the electron-hole recombine causes spontaneous emission & diode acts as a LED.

When the current is increased and reaches a threshold value, population inversion is acheived in the depletion region due to large concentration of electrons in the conduction band and holes in the valence band. The narrow region where the population inversion is acheived becomes

Forward biased



active region and lasing action takesplace. The Forward bias applied to the junction does the pumping mechanism & hence population inversion is obtained. (9)

The photon travelling in the direction along the resonant cavity stimulates recombination of electron-hole pairs due to which intensity of light builds up & hence emerged as a laser light across the depletion region.

The semiconductor laser is compact and efficient and have low power consumption. The drawback of this laser is less mono-chromatic and more divergent compared to other lasers.

The energy gap of GaAs is $E_g = 1.4\,eV$.

$\therefore$ The wavelength emitted by the diode laser is

$$\lambda = \frac{hc}{E_g} = \frac{6.625\times10^{-34}\times 3\times10^{8}}{1.4\times1.6\times10^{-19}}$$

$$\boxed{\lambda \simeq 8400\,\mathring{A}}$$ (Far red color).

Applications of laser next page.

P.T.O

Chapter 2

Semiconductor Devices

Syllabus
<u>**Semiconductor Devices**</u>
Photo-diode and Power responsivity, Construction and working of Semiconducting Laser, Four probe method to determine resistivity, Phototransistor.

2.1 Photodiode

A photodiode is a PN junction diode that converts light into electrical current. It is also known as a photodetector or a photosensor. It operates based on the principle of the photovoltaic effect. When light (photons) with sufficient energy strikes the semiconductor material of the photodiode, it can excite electrons from the valence band to the conduction band, creating electron-hole pairs. The generated electron-hole pairs are then separated by the built-in electric field within the photodiode, resulting in a flow of current.

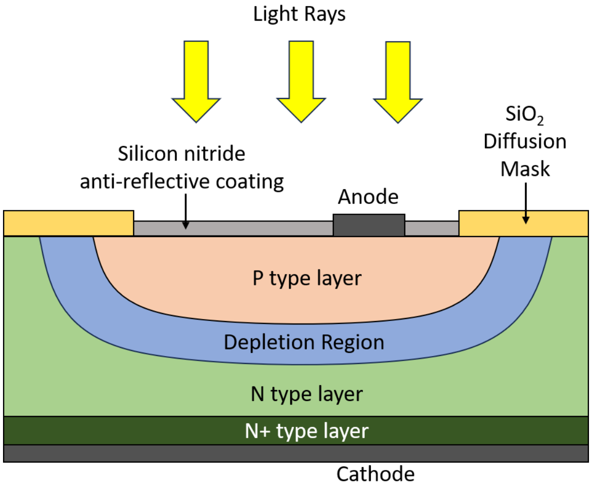


Figure 2.1: Photodiode Cross section

Construction

The typical construction of a photodiode is illustrated in figure 2.1. During the formation of the diode, electrons from the N type are attracted into the P type material and holes from the P type are attracted into the N type layer, resulting in the removal of free charge carriers close to the PN junction, so creating a depletion layer.

The (light-facing) top of the diode is protected by a layer of silicon dioxide (SiO_2) in which there is a window for light to shine on the semiconductor. This window is coated with a thin anti-reflective layer of silicon nitride (SiN) to allow maximum absorption of light. Also, an anode connection of aluminium is provided to the P type layer. Beneath the N type layer is a more heavily doped N+ layer to provide a low resistance connection to the cathode.

For a diode operating in Photoconductive Mode (presence of external voltage), it is usual to use reverse bias by applying a DC voltage to make the cathode more positive than the anode. The reverse bias causes the potential across the depletion region to increase and the width of the depletion region to increase. This is ideal for creating a large area to absorb the maximum amount of photons. And also, because the P and N layers with the depletion layer between them effectively form a capacitor, widening the depletion layer reduces the capacitance of the PN junction and increases the maximum frequency at which the diode can operate; a desirable property, especially in photodiodes that operate as digital information receivers.

Working

When the surface of the photodiode is illuminated, photons are absorbed within the diode (mainly in the depletion layer) which energises the electrons in the valence band to jump to the conduction band. This leaves behind positively charged holes in the valence band, resulting in the formation of electron-hole pairs in the depletion layer.

(Some electron-hole pairs are also produced in the P and N layers. But apart from those produced in the diffusion region, most will re-combine within the P and N materials as heat.)

The electrons in the depletion layer are then swept towards the positive potential on the cathode, and the holes swept towards the negative potential on the anode giving rise to current.

2.1.1 V-I characteristics of a photodiode

Photodiode operates in the reverse bias condition. Reverse voltages are plotted along X axis in volts and reverse current are plotted along Y-axis in microampere as seen in figure 2.2. Reverse current does not depend on reverse voltage. When there is no light illumination, reverse current will be almost zero. The minimum amount of current present is called as *Dark Current*. Once when the light illumination increases, reverse current also increases linearly.

2.1.2 Responsivity of a photodiode

The responsivity of a photodiode is the ratio of generated photocurrent and incident optical power, determined in the linear region of response.

Responsivity is a function of the wavelength of the incident radiation and of the sensor's properties, such as the bandgap of the material of which the photodetector is made. One expression for the responsivity R of a photodiode in which an optical signal is converted into a photocurrent is

$$R = \eta \frac{e}{h\nu}$$

where η is a dimensionless quantity called the quantum efficiency (the conversion efficiency of photons to electrons) of the photodiode for a given wavelength, e is the electron charge, ν is the frequency of the optical signal, and h is Planck's constant.

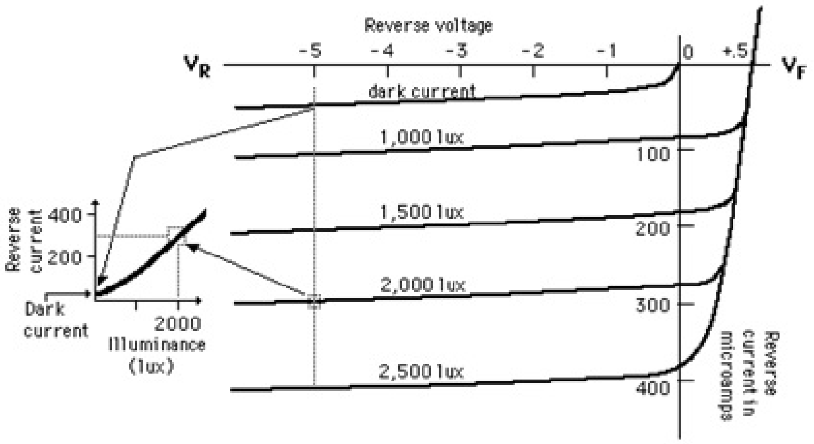


Figure 2.2: V-I characteristics

This expression can also be written in terms of λ, the wavelength of the optical signal, as

$$R = \eta \frac{e\lambda}{hc}$$

Responsitivity has units of coulombs per joule (C/J) or more commanly amperes per watt (A/W).

Applications

1. **Optical Communications:** Photodiodes are used in fiber optic communication systems to convert optical signals into electrical signals for processing.

2. **Photometry:** They are used in light meters and exposure meters in photography to measure the intensity of light.

3. **Remote Sensing:** They are used in remote sensing applications to detect and measure light reflected from the Earth's surface.

4. **Barcode Readers:** They are used in barcode scanners to detect the reflection of light from the barcode and convert it into an electrical signal.

5. **Smoke Detection:** They are used in smoke detectors to detect the presence of smoke particles, which scatter light onto the photodiode.

6. **Medical Instruments:** They are used in medical instruments for measuring oxygen saturation levels in blood and for non-invasive glucose monitoring.

7. **Security Systems:** They are used in security systems for detecting motion and light changes.

8. **Automotive:** They are used in automotive applications for ambient light sensing, automatic headlight control, and rain sensing.

9. **Consumer Electronics:** They are used in devices such as digital cameras, optical mice, and proximity sensors.

10. **Solar Energy:** They are used in solar cells to convert sunlight into electrical energy.

2.2 Phototransistor

A phototransistor is a light-controlled switch that switches a circuit and amplifies the current when exposed to light. It is a three-layer semiconductor device whose light-sensitive base is exposed. The light striking the base converts into a base current that amplifies the current between the emitter and collector proportional to the intensity of light. They are used for sensing light pulses of high speed and small magnitude.

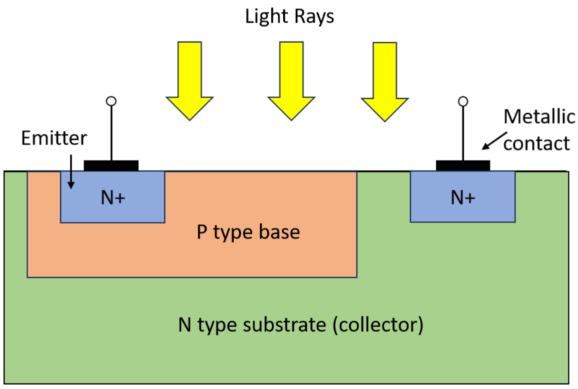


Figure 2.3: Phototransistor Cross section

Construction

When compared to normal transistor, in phototransistor the base and collector area is large (see figure 2.3). The base area is increased to increase the amount of incident light radiation. Earlier it was made up of single semiconductor material like silicon or germanium. Recently phototransistors are made up of Gallium and Arsenic to obtain higher efficiency. Phototransistor is placed inside a metallic case and a lens is kept at the top of the case to absorb the incident radiation.

Working

Phototransistor operates just like any normal bipolar transistor except for the fact that the base current is generated by a light source instead of a voltage source. The base current is generated on the principle of the photovoltaic effect. The base current is then amplified by the transistor action. Therefore the phototransistor is 100 times more sensitive than the photodiode.

When biasing, the collector is kept at a higher voltage with respect to the emitter in NPN phototransistor while in PNP, the collector is at a lower voltage with respect to the emitter. The collector to the base junction is reverse biased. The base terminal is kept open or not connected.

Under no light conditions, there is a small reverse saturation current or leakage current called dark current that is directly proportional to the temperature as in photodiodes.

Applications

1. **Light Sensors:** Phototransistors are commonly used as light sensors in applications such as automatic streetlights, camera light meters, and ambient light sensors in electronic devices. They can detect changes in light intensity and trigger corresponding actions or adjustments.

2. Barcode Readers: Phototransistors are used in barcode readers to detect the reflected light from the barcode pattern. They can convert the light pattern into electrical signals, which are then decoded to read the information stored in the barcode.
3. Smoke Detectors: Phototransistors are used in smoke detectors to detect the presence of smoke particles in the air. When smoke particles block the light falling on the phototransistor, it triggers an alarm, indicating a possible fire.
4. Optical Switches: Phototransistors are used in optical switches to control the flow of light in optical communication systems. They can be used to switch between different optical paths or to modulate the intensity of light signals.
5. Industrial Automation: Phototransistors are used in industrial automation systems for tasks such as object detection, position sensing, and speed sensing. They are used in conjunction with light barriers or sensors to detect the presence or absence of objects on a conveyor belt or in a production line.

2.4 Four-probe method

The four-probe apparatus is one of the standard and most widely used apparatus for the measurement of resistivity of semiconductors.

The apparatus consists of four equally spaced tungsten metal tips with finite radius (see figure 2.6). Each tip is supported by springs on the end to minimize sample damage during probing. The four metal tips are part of an auto-mechanical stage which travels up and down during measurements. A high impedance current source is used to supply current through the outer two probes, a voltmeter measures the voltage across the inner two probes to determine the sample resistivity. Typical probe spacing is around 2 mm. These inner probes draw no current because of the high input impedance voltmeter in the circuit. Thus unwanted voltage drop at point 2 and point 3 caused by contact resistance between probes and the sample is eliminated from the potential measurements.

In order to use this four-probe method in germanium crystals or slices it is necessary to assume that:

- The resistivity of the material is uniform in the area of measurement.
- A non conducting boundary is produced when the surface of the crystal is in contact with an insulator.

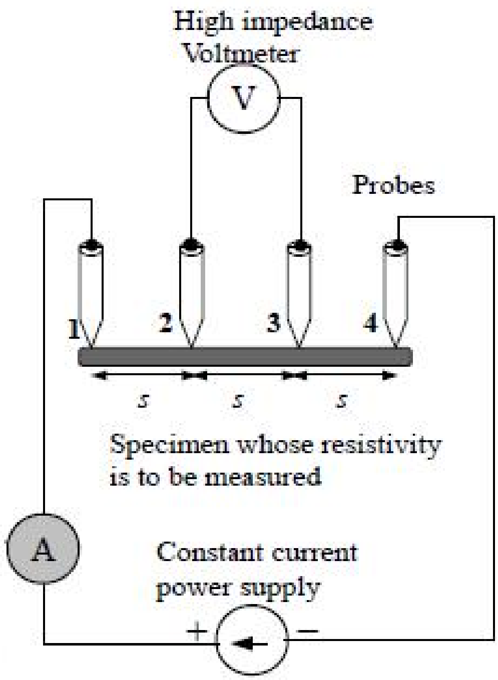


Figure 2.6: Four probe method apparatus

For each case it is assumed that the probes are equally spaced (spacing = s).

Case 1: Resistivity Measurements on a Large Sample

We assume that the metal tip is infinitesimal and sample is semi infinite in lateral dimensions. For bulk samples where the sample thickness, $w >> s$, the probe spacing, we assume a spherical protrusion of current emanating from the outer probe tips. The resistivity is computed to be

$$\rho_0 = \frac{V}{I} \times 2\pi s$$

where,
V = floating potential difference between the inner probes (volt)
I = current through the outer pair of probes (ampere)
s = spacing between point probes (meter)
ρ_0 = resistivity (ohm meter)

Case 2: Resistivity Measurements on a Thin Slice

For the case of a nonconducting bottom on a slice the resistivity is computed from

$$\rho = \frac{\rho_0}{f\left(\frac{w}{s}\right)}$$

where, $f\left(\frac{w}{s}\right) = \frac{2s}{w}\ln(2)$.

Temperatue dependence of resistivity

The resistivity ρ of a semiconductor depends on the temperatue T as follows

$$\rho = A \exp\left(\frac{E_g}{2k_B T}\right)$$

where A is a constant, E_g is the energy gap of the semiconductor.

Applications

1. **Semiconductor Research**: The four-probe method is widely used in the development of new semiconductor devices, as it allows researchers to accurately measure the resistivity of semiconductor materials and thin films.
2. **Quality Control**: In semiconductor manufacturing, the four-probe method is used in quality control processes to ensure that materials meet the required resistivity specifications.
3. **Thin Film Technologies**: The four-probe method is used to characterize the electrical properties of thin films, which are used in a variety of applications including solar cells, displays, and sensors.
4. **Fundamental Semiconductor Properties**: The method is also used in fundamental research to study the electrical properties of semiconductors and to investigate new materials with potential applications in electronics.

2.5 Model and Previous Year Questions

Q 1. Explain how the resistivity of a semiconductor is determined using four probe method? Mention any two applications of four probe method.

Q 2. Explain the construction and working of photodiode. Discuss the power responsivity in a photodiode.

Q 3. Describe the construction and working of semiconductor laser with energy level diagram.

Q 4. Explain the construction and working of phototransistor and also mention any two applications.